

Tech Note T27: Simple noise models

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In this little pedagogic guide, I review a few simple but key noise models. In particular, I focus on global depolarizing and bit-flip processes. I'll sketch out various ways to see them and connect the dots to key concepts like average gate fidelity, entanglement fidelity. Plus, we'll see how they fit with that fancy sparse-Pauli-Lindblad noise model, which we use in tricks like probabilistic error cancelation (PEC) and twirled readout error mitigation (T-REX). Let's explore!

27	Simple noise models	1
27.1	Depolarizing noise	1
27.1.1	Single qubit	1
27.1.2	n -qubit depolarizing channel	4
27.2	Bit flip noise channel	6
27.2.1	Single qubit	6
27.3	Gate fidelity	7
27.3.1	State fidelity	7
27.3.2	Average gate fidelity and entanglement fidelity	7
27.4	Fidelity between a cNOT and a (Z,X) random process	9
27.5	Pauli conjugated by Pauli [MOVE]	10
27.5.1	Conjugation of target Pauli by ctrl-Pauli	11

27.1 Depolarizing noise

This is Mother Nature's sneaky way of saying, "I'm going to randomly destroy your system a little bit." Specifically, with a certain probability p , she decides to completely erase all information in the system. But with a probability $1 - p$, she lets the qubit pass undisturbed, thinking, "Next time!"

27.1.1 Single qubit

Let's start with the single qubit example. The depolarizing channel Λ can be expressed in equivalent ways, each defined with its parameter highlighted in blue and explained in depth below the equations:

$$\begin{aligned}\Lambda(\rho) &= (1 - p) I\rho I + \frac{p}{4} (I\rho I + X\rho X + Y\rho Y + Z\rho Z) , & 0 \leq p \leq 1, \quad \chi \text{ basis} & \quad (1) \\ &= \left(1 - \frac{3}{4}p\right) I\rho I + \frac{p}{4} (X\rho X + Y\rho Y + Z\rho Z) , & \text{simple} & \\ &= (1 - p) I\rho I + p \left[\frac{1}{2} I \text{Tr}(I\rho) \right] , & \hat{\pi} := \frac{1}{2} I \text{Tr}(I\rho) , \quad \text{simple state} & \\ &= (1 - p') I\rho I + \frac{p'}{3} (X\rho X + Y\rho Y + Z\rho Z) , & p' := \frac{3}{4}p , \quad \text{alt simple} & \\ \Lambda \cdot &= \frac{1}{2} |I\rangle\langle I| \cdot + \frac{1}{2} f (|X\rangle\langle X| + |Y\rangle\langle Y| + |Z\rangle\langle Z|) \cdot , & f := 1 - p , \quad \text{PTM basis} & \\ &= \exp [\lambda (\mathcal{D}[X] \cdot + \mathcal{D}[Y] \cdot + \mathcal{D}[Z] \cdot)] , & \lambda := -\frac{1}{4} \ln(f) , \quad \text{generator basis} & \quad (2)\end{aligned}$$

Explanations:

- p the is the depolarizing probability, $0 \leq p \leq 1$ (actually for a valid CPTP channel $0 \leq p \leq \frac{4}{3}$, see below), to get scrambled to a maximally mixed state $\hat{\pi} := \frac{1}{2} \hat{I}$.
 - $p = 0$ means there is no error

- $p = 1$ means the output is the maximally depolarizes the state always, in which case $\mathcal{D}(\rho) = \frac{1}{4}(I\rho I + X\rho X + Y\rho Y + Z\rho Z) = |I\rangle\rangle\langle\langle I| = I \text{Tr}(I \cdot)$. Note that I use unnormalized Pauli super-kets $|P_a\rangle\rangle = P_a$.
- p' is the ‘probability’ that the state remaining intact, while with probability $p'/3$ the state is subjected to a bit flip X , phase flip Z , or both (phase and bit flip) Y error, $0 \leq p' \leq \frac{4}{3}$.
- f is the Pauli fidelity of the channel for a given Pauli component. They are all equal.
- λ is the generator of the noise, $\lambda := -\frac{1}{4} \ln(f) = \frac{1}{4} \ln\left(\frac{1}{1-p}\right)$, equally distributed for each Pauli, where $\mathcal{D}[A](\rho) := A\rho A^\dagger - \frac{1}{2}(A^\dagger A\rho + \rho A^\dagger A)$. There is no $\mathcal{D}[I] \cdot$, because it won’t show up in the exponentiation, no matter what the coefficient is.

The basic idea is that the error rate on each Pauli term is identical except for the error on the identity Pauli, which must always be conserved for $\text{Tr}(\rho) = 1$, i.e., $|I\rangle\rangle\langle\langle I|$. It is easiest recognized in the Pauli transfer matrix basis.

Conversions

Row in column	p	f	λ
p	$\cdot$	$p = 1 - f$	$p = 1 - \exp(-4\lambda)$
f	$f := 1 - p$	$\cdot$	$f = \exp(-4\lambda)$
λ	$\lambda = -\frac{1}{4} \ln(1 - p)$	$\lambda := -\frac{1}{4} \ln(f)$	$\cdot$

Fidelity The *average gate or identity process fidelity* as defined in Sec. 41 for the single qubit depolarizing channel is

$$F_{\text{avg}}(\Lambda) = \int d\psi \langle\langle \rho_\psi | \Lambda | \rho_\psi \rangle\rangle \quad (3)$$

$$= 1 - \frac{p}{2} . \quad (4)$$

The worst case fidelities are $F = \frac{1}{2}$ for $p = 1$ and $F = \frac{1}{3}$ for $p = 4/3$. The entanglement fidelity, see Eq. (43), is $F_e = 1 - \frac{3}{4}p$.

Positivity For Λ to be a valid channel we require that it is a completely positive trace-preserving (CPTP) map. From the PTM representation above, we see that Λ is TP for any p . However, for CP, we need to check that the Choi matrix $J(\Lambda)$ is positive semidefinite. The eigenvalues of $J(\Lambda)$ are $\{\frac{1}{2}(4 - 3p), \frac{p}{2}, \frac{p}{2}, \frac{p}{2}\}$. From this we can drive the bound on p :

$$0 \leq p \leq \frac{4}{3} .$$

This tells us that p can be greater than 1 and the channel would still be valid. At the bound $p = \frac{4}{3}$, the PTM and process χ matrices in the Pauli basis are:

$$\begin{aligned} \text{PTM}(\Lambda) &= \begin{matrix} & \begin{matrix} I & X & Y & Z \end{matrix} \\ \begin{matrix} I \\ X \\ Y \\ Z \end{matrix} & \begin{pmatrix} 1 & & & \\ & 1-p & & \\ & & 1-p & \\ & & & 1-p \end{pmatrix} \end{matrix} = \begin{matrix} & \begin{matrix} I & X & Y & Z \end{matrix} \\ \begin{matrix} I \\ X \\ Y \\ Z \end{matrix} & \begin{pmatrix} 1 & & & \\ & -\frac{1}{3} & & \\ & & -\frac{1}{3} & \\ & & & -\frac{1}{3} \end{pmatrix} \end{matrix} \\ \chi(\Lambda) &= \begin{matrix} & \begin{matrix} I & X & Y & Z \end{matrix} \\ \begin{matrix} I \\ X \\ Y \\ Z \end{matrix} & \begin{pmatrix} 1 - \frac{3}{4}p & & & \\ & \frac{p}{4} & & \\ & & \frac{p}{4} & \\ & & & \frac{p}{4} \end{pmatrix} \end{matrix} = \begin{matrix} & \begin{matrix} I & X & Y & Z \end{matrix} \\ \begin{matrix} I \\ X \\ Y \\ Z \end{matrix} & \begin{pmatrix} 0 & & & \\ & \frac{1}{3} & & \\ & & \frac{1}{3} & \\ & & & \frac{1}{3} \end{pmatrix} . \end{matrix} \end{aligned}$$

Connection to product channels As an aside, to connect to the weight introduced in our paper (see Probabilistic error cancellation with sparse Pauli-Lindblad models on noisy quantum processors), observe that in Eq. (2), the generator terms commute $[\mathcal{D}[P_a], \mathcal{D}[P_b]] = 0$ for different Paulis P_a and P_b . It follows that we can factorize the exponential of the sum into a product of exponentials. We can then factorize the channel into a product of channel, each of which is given by a single Pauli, and we can write out the product explicitly:

$$\Lambda = \exp[\lambda(\mathcal{D}[X] \cdot + \mathcal{D}[Y] \cdot + \mathcal{D}[Z] \cdot)] \quad (5)$$

$$= \exp[\lambda \mathcal{D}[X]] \exp[\lambda \mathcal{D}[Y]] \exp[\lambda \mathcal{D}[Z]] \quad (6)$$

$$= \Lambda_X \Lambda_Y \Lambda_Z \quad (7)$$

$$= (w\mathcal{I} + (1-w)\mathcal{X})(w\mathcal{I} + (1-w)\mathcal{Y})(w\mathcal{I} + (1-w)\mathcal{Z}) , \quad (8)$$

$$w := e^{-\lambda} \cosh(\lambda) = \frac{1}{2} (1 + e^{-2\lambda}) = \frac{1}{2} (1 + \sqrt{1-p}) , \quad (9)$$

$$1-w = e^{-\lambda} \sinh(\lambda) = \frac{1}{2} (1 - e^{-2\lambda}) = \frac{1}{2} (1 - \sqrt{1-p}) , \quad (10)$$

$$\Lambda_a := \exp[\lambda \mathcal{D}[P_a]] = w\mathcal{I} + (1-w)\mathcal{P}_a , \quad (11)$$

where we have introduced the weights or probabilities of no error per term w and the sub-channel decompositions Λ_a , which all commute. In Eq. (8), we see that the noise channel is factorized product of independent noise channels.

$$\text{---} \boxed{\Lambda} \text{---} = \text{---} \boxed{\Lambda_X} \text{---} \boxed{\Lambda_Y} \text{---} \boxed{\Lambda_Z} \text{---}$$

27.1.2 n -qubit depolarizing channel

Generalizing Eq. (1), the global depolarizing channel completely randomizes the state to the global mixed state $\hat{I}/\dim(\mathcal{H})$ with probability p , $0 \leq p \leq 1$ (more generally, for a valid CPTP: $0 \leq p \leq 1 + \frac{1}{d^2 - 1}$), where $\dim(\mathcal{H}) = d = 2^n$, and with complementary probability $1 - p$ it leave it untouched:

$$\Lambda(\rho) = (1 - p)\rho + p\frac{1}{d}\hat{I}\text{Tr}(\rho) \quad \text{simple} \quad (12)$$

$$= (1 - p)I\rho I + \frac{p}{d^2} \sum_{a=0}^{4^n-1} P_a \rho P_a, \quad \chi \text{ basis} \quad (13)$$

$$\Lambda = \frac{1}{d} \left(|I\rangle\rangle\langle\langle I| + f \sum_{a=1}^{4^n-1} |P_a\rangle\rangle\langle\langle P_a| \right), \quad \text{PTM basis} \quad (14)$$

$$= \exp \left[\lambda \sum_{a=1}^{4^n-1} \mathcal{D}[P_a] \right] \quad \text{generator basis} \quad (15)$$

$$= \prod_{a=1}^{4^n-1} (w\mathcal{I} + (1 - w)\mathcal{P}_a) \quad \text{product channels} \quad (16)$$

where all the Pauli fidelities f are equal (aside from that of the identity), $\langle\langle P_a | P_b \rangle\rangle = d\delta_{ab}$, with

$$f = 1 - p, \quad p = 1 - f, \quad (17)$$

$$f = \exp(-d^2\lambda), \quad \lambda = -\frac{1}{d^2} \ln(f), \quad (18)$$

and the sampling overhead factor

$$\gamma = \exp \left(\sum_k 2\lambda_k \right) = f^{-2\frac{d^2-1}{d^2}} \approx \frac{1}{f^2} = \frac{1}{(1-p)^2}. \quad (19)$$

Note that $\dim(L(\mathcal{H})) = d^2$. Note that Λ is a *unitarily invariant* superoperator, $\mathcal{U}\Lambda\mathcal{U}^\dagger = \Lambda$, i.e. it is unchanged under unitary rotations $\mathcal{U}$. This implies it is a fixed point of Haar average measures and global twirling. The average gate fidelity, see Sec. 27.3, is

$$F_{\text{avg}}(\Lambda) = 1 - p + \frac{p}{d}, \quad (20)$$

where we recognize that $p = 0 \implies F_{\text{avg}} = 1$ and $p = 1 \implies F_{\text{avg}} = \frac{1}{d} = \frac{1}{2^n}$, which is just $\text{Tr}(\hat{\pi}\rho) = \frac{1}{2^n}$, where the maximally mixed state is $\hat{\pi} = \frac{I}{2^n}$. When we want to convert a measured fidelity to a simulation probability of an error, the probability of a depolarizing error is

$$p = \frac{d}{d-1}(F_{\text{avg}} - 1), \quad (21)$$

where for a two-qubit channel, $d = 4$.

Two-qubit depolarizing noise and gate fidelity Since the gate fidelity of two qubits is often measured and quoted as $F \sim \mathcal{O}(1)$, and we want to convert it to a depolarizing error rate p , we can use Eq. (21): $p = \frac{4}{3}(1 - F)$.

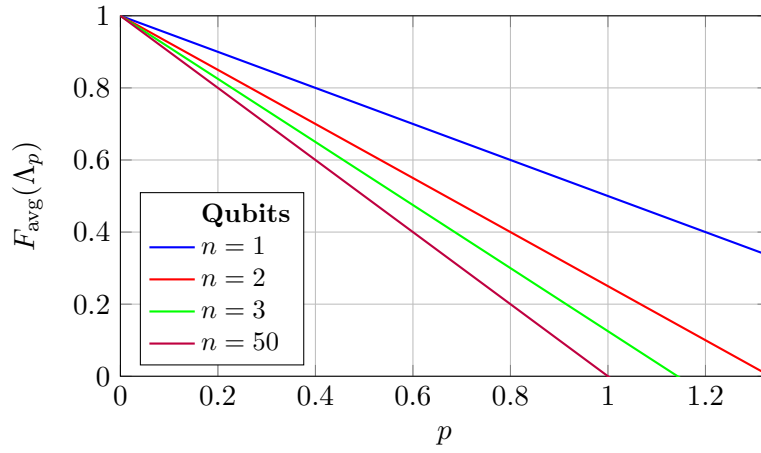


Figure 1: Average gate fidelity for depolarizing channel as a function of the probability of an error p .

Connection to product channels We can generalize the results of Sec. 27.1.1 straightforward, recalling Eq. (6), starting from Eq. (15),

$$\Lambda = \exp \left[\lambda \sum_{a=1}^{4^n-1} \mathcal{D}[P_a] \right] \quad (22)$$

$$= \prod_{a=1}^{4^n-1} \exp [\lambda \mathcal{D}[P_a]] \quad (23)$$

$$= \prod_{a=1}^{4^n-1} \Lambda_a \quad (24)$$

$$\Lambda_a := \exp [\lambda \mathcal{D}[P_a]] \quad (25)$$

$$= w\mathcal{I} + (1-w)\mathcal{P}_a \quad (26)$$

$$w := e^{-\lambda} \cosh(\lambda) = \frac{1}{2} (1 + e^{-2\lambda}) = \frac{1}{2} (1 + \sqrt{1-p}) , \quad (27)$$

$$1-w = e^{-\lambda} \sinh(\lambda) = \frac{1}{2} (1 - e^{-2\lambda}) = \frac{1}{2} (1 - \sqrt{1-p}) , \quad (28)$$

where Λ_a and w are exactly as in the one-qubit case, even though they are now based on n -qubit Pauli operators P_a .

27.2 Bit flip noise channel

A bit-flip error is analogous to the classical bit-flip error where a bit's value is flipped: 0 becomes 1 and 1 becomes 0. In quantum mechanics, the bit-flip error applies to qubits and results in the flipping of a quantum state.

27.2.1 Single qubit

For an error probability $0 \leq p \leq 1$, $d = 2$, $f := 1 - 2p$, $\lambda := -\frac{1}{2} \ln(f)$, where we

$$\Lambda(\cdot) = (1 - p) I \cdot I + p X \cdot X \quad (29)$$

$$= \frac{1}{d} [|I\rangle\langle I| + |X\rangle\langle X| + f(|Y\rangle\langle Y| + |Z\rangle\langle Z|)] \cdot \quad (30)$$

$$= \exp[\lambda \mathcal{D}[X]] \cdot \quad (31)$$

Note that $1 - p = \frac{1+f}{2}$ and $p = \frac{1-f}{2}$, and that the sampling overhead is

$$\gamma = \left\lceil \frac{1+f^{-1}}{2} \right\rceil + \left\lceil \frac{1-f^{-1}}{2} \right\rceil = \exp(2\lambda) = \frac{1}{f} \cdot \quad (32)$$

The conversion among all the representations is summarized in the following table:

Row in column	p	f	λ	γ
p	.	$p = \frac{1-f}{2}$	$p = \frac{1}{2} (1 - e^{-\lambda/2})$	$p = \frac{1-\gamma^{-1}}{2}$
f	$f := 1 - 2p$	.	$f = e^{-\lambda/2}$	$f = \frac{1}{\gamma}$
λ	$\lambda = -\frac{1}{2} \ln(1 - 2p)$	$\lambda := -\frac{1}{2} \ln(f)$	.	$\lambda := +\frac{1}{2} \ln(\gamma)$
γ	$\gamma = \frac{1}{1-2p}$	$\gamma := \frac{1}{f}$	$\gamma = e^{+\lambda/2}$	.

Average gate fidelity Derived as shown in Sec. 27.3, the average gate fidelity is

$$F_{\text{avg}}(\Lambda) = 1 - \frac{2}{3}p \cdot, \quad (33)$$

or $p = \frac{3}{2}(F - 1)$. For $p = 1$, $F = \frac{1}{3}$, which you will note is higher than the $p = 1$ fidelity of the depolarizing channel, which is $\frac{1}{4}$. In this sense it is clear that the depolarizing channel induced more errors.

Inverse noise channel For the inverse noise channel Λ^{-1} , just replace $f \mapsto f^{-1}$ and $p \mapsto q = -p/(1 - 2p)$ and $\lambda \mapsto -\lambda$.

Positivity From the Choi matrix $J(\Lambda)$ eigenvalues, one can verify that $0 \leq p \leq 1$, and p cannot be larger than one for a CPTP map.

27.3 Gate fidelity

If an operation (let's call it a gate) promises to take our qubit from State A to State B, the fidelity tells us how reliably it does so. A fidelity of 1 means perfect reliability, while a value less than 1 indicates some error. Gate fidelity is based on a measure of the similarity between the two output quantum states of the ideal and noisy processes. To define it, we start with the fidelity of two quantum states.

27.3.1 State fidelity

To connect to experimental measurements, let us decompose the quantum states in the standard Pauli basis. The set of all Pauli operators on n qubits $\{I, X, Y, Z\}^{\otimes n}$ forms an orthogonal Hermitian operator basis. The inner product in operator space $L(\mathcal{H})$ between two Pauli operators $P_i, P_j \in \mathcal{L}(\mathcal{H})$ is $\langle P_i, P_j \rangle = \text{Tr}(P_i P_j) = d \delta_{ij}$, where the dimension of the pure state Hilbert space $d := \dim \mathcal{H} = 2^n$. In terms of this basis, any quantum state $\rho \in D(\mathcal{H})$, can be decomposed into

$$\rho = \sum_{i=0}^{4^n-1} \frac{\langle P_i, \rho \rangle}{\langle P_i, P_i \rangle} P_i = \frac{1}{d} \sum_{i=0}^{4^n-1} \rho_i P_i, \quad \rho_i := \langle P_i, \rho \rangle = \text{Tr}(P_i \rho),$$

Concretely, the **Uhlmann–Jozsa state fidelity** between two quantum state $\rho, \sigma \in D(\mathcal{H})$ is

$$F(\rho, \sigma) = \left[\text{Tr} \left(\sqrt{\sqrt{\rho} \sigma \sqrt{\rho}} \right) \right]^2 \quad (34)$$

$$= \text{Tr}(\rho \sigma)^2, \quad \text{general, see Muller}, \quad (35)$$

$$= \langle \rho, \sigma \rangle^2 \quad (36)$$

$$= \langle \langle \rho | \sigma \rangle \rangle^2 \quad (37)$$

$$= \langle \phi | \rho | \phi \rangle, \quad \text{for pure } \sigma = |\phi\rangle \langle \phi|, \quad (38)$$

$$= \text{Tr}[\rho \sigma], \quad \text{for pure } \rho, \sigma, \quad (39)$$

$$= |\langle \psi | \phi \rangle|^2. \quad (40)$$

For pure states, projectors, $\sqrt{\rho} = \rho$, $\rho = |\psi\rangle \langle \psi|$ and $\sigma = |\phi\rangle \langle \phi|$. So, the Uhlmann's fidelity between a mixed state ρ and a pure state $\sigma = |\phi\rangle \langle \phi|$ is just the probability of measuring the state ρ in the state $|\phi\rangle$. The second line is shown in detail in Ref. (2023 Muller - A Simplified Expression for Quantum Fidelity), it does *not* require that ρ and σ commute, but is general. Thus in general, the Fidelity is just the inner product of ρ and σ squared. Note that $\sqrt{\rho}$ denotes the *positive* square root of ρ .

27.3.2 Average gate fidelity and entanglement fidelity

The **average gate fidelity** can be thought of as an average over all possible input states of the fidelity between the state produced by the actual operation and the state produced by the ideal operation. For a noise channel Λ the average fidelity then is, given $\rho_\psi = |\psi\rangle \langle \psi|$ and over a uniform Haar average,

$$\begin{aligned} F_{\text{avg}}(\Lambda) &= \mathbb{E}_\psi [F(\rho_\psi, \Lambda(\rho_\psi))] \\ &= \int d\psi \langle \psi | \Lambda(|\psi\rangle \langle \psi|) | \psi \rangle \\ &= \int d\psi \langle \langle \rho_\psi | \Lambda | \rho_\psi \rangle \rangle. \end{aligned} \quad (41)$$

In terms of the Pauli transfer matrix T_Λ of the channel Λ , it can be computed

$$F_{\text{avg}}(\Lambda) = \frac{\frac{1}{d} \text{Tr}(T_\Lambda) + 1}{d + 1}, \quad (42)$$

where $d = \dim \mathcal{H} = 2^n$.

It can be related to the **entanglement fidelity** $F_e(\Lambda)$, (see A simple formula for the average gate fidelity of a quantum dynamical operation, Eq. (3)),

$$F_{\text{avg}}(\Lambda) = \frac{d F_e(\Lambda) + 1}{d + 1}. \quad (43)$$

Entanglement fidelity provides a measure of how closely a given quantum operation or channel preserves the entanglement of an input state. It's particularly useful for assessing the quality of quantum channels and quantum operations in terms of their ability to preserve quantum features like entanglement. Why is it useful? For example, for a composite channel $\mathcal{A} = \mathcal{A}_1(p) \mathcal{A}_2(q)$ acting with say amplitude damping noise with probability p on qubit 1 and probability q on qubit 2, the average gate fidelities do not multiply, but the entanglement fidelities do $F_e(\mathcal{A}) = F_e(\mathcal{A}_1) F_e(\mathcal{A}_2)$.

Comparing an ideal unitary Consider the case in which we want to implement an ideal gate $\mathcal{U} \cdot := U \cdot U^\dagger$. However, instead we can implement only a noisy gate $\tilde{\mathcal{U}} := \mathcal{U}\Lambda$, where Λ is some effective noise channel. What is the gate fidelity of noisy $\tilde{\mathcal{U}}$ relative to the ideal $\mathcal{U}$. The gate fidelity for a given pure state $\rho = |\phi\rangle\langle\phi|$ between the two channels is

$$F(\mathcal{U}, \tilde{\mathcal{U}}; \rho) = \left[\text{Tr} \left[\sqrt{\sqrt{\mathcal{U}(\rho)} \tilde{\mathcal{U}}(\rho) \sqrt{\mathcal{U}(\rho)}} \right] \right]^2 \quad (44)$$

$$= \text{Tr} [\mathcal{U}(\rho) \tilde{\mathcal{U}}(\rho)] = \langle\langle \rho | \mathcal{U}^\dagger \tilde{\mathcal{U}} | \rho \rangle\rangle \quad (45)$$

$$= \text{Tr} [\mathcal{U}(\rho) \mathcal{U}(\Lambda\rho)] \quad (46)$$

$$= \text{Tr} [U \rho U^\dagger U \Lambda(\rho) U^\dagger] \quad (47)$$

$$= \text{Tr} [\rho \Lambda(\rho)] \quad (48)$$

$$= \text{Tr} [|\phi\rangle\langle\phi| \Lambda(|\phi\rangle\langle\phi|)] \quad (49)$$

$$= \langle\langle \rho | \Lambda | \rho \rangle\rangle \quad (50)$$

To obtain the **average gate fidelity** then, with $\rho_\psi = |\psi\rangle\langle\psi|$, and $d = \dim \mathcal{H} = 2^n$,

$$F_{\text{avg}}(\mathcal{U}, \tilde{\mathcal{U}}) = \mathbb{E}_\psi [F(\mathcal{U}, \tilde{\mathcal{U}}; \rho_\psi)] \quad (51)$$

$$= \int d\psi F(\mathcal{U}, \tilde{\mathcal{U}}; \rho_\psi) \quad (52)$$

$$= \int d\psi \text{Tr} [\mathcal{U}(\rho) \tilde{\mathcal{U}}(\rho)] \quad (53)$$

$$= \int d\psi \langle\langle \rho | \mathcal{U}^\dagger \tilde{\mathcal{U}} | \rho \rangle\rangle \quad (54)$$

$$= \int d\psi \langle\langle \rho | \mathcal{U}^\dagger \mathcal{U} \Lambda | \rho \rangle\rangle \quad (55)$$

$$= \int d\psi \langle\langle \rho | \Lambda | \rho \rangle\rangle \quad (56)$$

$$= \int d\psi \text{Tr} [|\phi\rangle\langle\phi| \Lambda(|\phi\rangle\langle\phi|)] \quad (57)$$

$$= \text{Tr} \left[\int d\psi |\phi\rangle\langle\phi| \Lambda(|\phi\rangle\langle\phi|) \right] \quad (58)$$

$$= \frac{\frac{1}{d} \text{Tr}(\Lambda) + 1}{d + 1} \quad (59)$$

If you want to compare quantum channels that are not unitary, you will need the inverse. What if the channel is singular

27.4 Fidelity between a cNOT and a (Z,X) random process

To understand possible failure modes of the two-qubit CNOT gate teleportation given the dynamic circuit depicted in Fig. 1, imagine that the conditional gates are implemented completely at random. That is with equal probability $\frac{1}{4}$, the two-qubit gate II , or ZI , or IX , or ZX is applied. To understand the average gate fidelity of a cNOT gate with this random process, we detail the following simple calculation using superoperator notation. We use the super-Dirac notation, where the super-ket $|P\rangle\rangle = P$ stands for an operator $P \in L(\mathcal{H})$ in the operator space on the state Hilbert space $\mathcal{H}$.

Let our target channel be the two-qubit CNOT gate, in superoperator language $c\mathcal{X} := \mathcal{U}(\cdot) := U \cdot U^\dagger$ with $U \in L(\mathcal{H})$ the two-qubit cNOT unitary on the two-qubit Hilbert space $\mathcal{H} = \mathbb{C}^{2^2}$. Let our noisy or random channel be, for the superoperator $\mathcal{S}[P] \cdot = P \cdot P^\dagger$,

$$\begin{aligned}\mathcal{P} &:= \frac{1}{4} (\mathcal{S}[II] + \mathcal{S}[ZI] + \mathcal{S}[IX] + \mathcal{S}[ZX]) \\ &= |II\rangle\rangle\langle\langle II| + |ZI\rangle\rangle\langle\langle ZI| + |IX\rangle\rangle\langle\langle IX| + |ZX\rangle\rangle\langle\langle ZX| ,\end{aligned}$$

which shows us the $\mathcal{P}$ channel is a diagonal stochastic Pauli channel with just 4 identity fidelities, and all others are zero. For the cNOT gate, keeping in mind $\text{cNOT}^\dagger = \text{cNOT}$:

$$\begin{aligned}c\mathcal{X} &= |II\rangle\rangle\langle\langle II| + |IX\rangle\rangle\langle\langle IX| + |IY\rangle\rangle\langle\langle ZY| + |IZ\rangle\rangle\langle\langle ZZ| + |XI\rangle\rangle\langle\langle XX| \\ &\quad + |XX\rangle\rangle\langle\langle XI| + |XY\rangle\rangle\langle\langle YZ| - |XZ\rangle\rangle\langle\langle YY| + |YI\rangle\rangle\langle\langle YX| \\ &\quad + |YX\rangle\rangle\langle\langle YI| - |YY\rangle\rangle\langle\langle XZ| + |YZ\rangle\rangle\langle\langle XY| + |ZI\rangle\rangle\langle\langle ZI| \\ &\quad + |ZX\rangle\rangle\langle\langle ZX| + |ZY\rangle\rangle\langle\langle IY| + |ZZ\rangle\rangle\langle\langle IZ| .\end{aligned}$$

Let the noise channel Λ be $\Lambda := \mathcal{U}^\dagger \tilde{\mathcal{U}} = c\mathcal{X}\mathcal{P}$ so that $\tilde{\mathcal{U}} = \mathcal{U}\Lambda$, which means it's PTM in super-Dirac notation is

$$\Lambda = c\mathcal{X}\mathcal{P} = \mathcal{P} = |II\rangle\rangle\langle\langle II| + |ZI\rangle\rangle\langle\langle ZI| + |IX\rangle\rangle\langle\langle IX| + |ZX\rangle\rangle\langle\langle ZX| .$$

$$\text{---} \boxed{\tilde{\mathcal{U}}} \text{---} = \text{---} \boxed{\Lambda} \text{---} \boxed{\mathcal{U}} \text{---}$$

Fidelity The fidelity for $\mathcal{U} = c\mathcal{X}$ and noisy $\tilde{\mathcal{U}} = \mathcal{P}$, with T_Λ the PTM of Λ and $d = 2^2$,

$$\begin{aligned}F_{\text{avg}}(\mathcal{U}, \tilde{\mathcal{U}}) &= \int d\psi \text{Tr} [\mathcal{U}(\rho) \tilde{\mathcal{U}}(\rho)] = \int d\psi \langle\langle \rho | \mathcal{U}^\dagger \tilde{\mathcal{U}} | \rho \rangle\rangle = \int d\psi \langle\langle \rho | \Lambda | \rho \rangle\rangle \\ &= \frac{\frac{1}{d} \text{Tr}(T_\Lambda) + 1}{d + 1} \\ &= \frac{\frac{4}{2^2} + 1}{2^2 + 1} = \frac{2}{5} = 0.4 .\end{aligned}$$

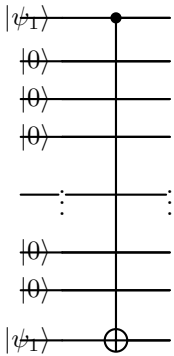
y The fidelity for $\mathcal{U} = c\mathcal{X}$ and noisy $\tilde{\mathcal{U}} = \mathcal{P}$, with T_Λ the PTM of Λ and $d = 2^2$,

$$\begin{aligned}F_{\text{avg}}(\mathcal{U}, \tilde{\mathcal{U}}) &= \int d\psi \text{Tr} [\mathcal{U}(\rho) \tilde{\mathcal{U}}(\rho)] \\ &= \int d\psi \langle\langle \rho | \Lambda | \rho \rangle\rangle \\ &= \frac{\frac{1}{d} \text{Tr}(T_\Lambda) + 1}{d + 1} \\ &= \frac{\frac{4}{2^2} + 1}{2^2 + 1} = \frac{2}{5} = 0.4 .\end{aligned}$$

Extra notes In terms of it's Pauli transfer matrices (PTM) and chi matrices:

$$c\mathcal{X}^{\text{PTM}} = \begin{pmatrix} & \text{II} & \text{IX} & \text{IY} & \text{IZ} & \text{XI} & \text{XX} & \text{XY} & \text{XZ} & \text{YI} & \text{YX} & \text{YY} & \text{YZ} & \text{ZI} & \text{ZX} & \text{ZY} & \text{ZZ} \\ \text{II} & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \text{IX} & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \text{IY} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ \text{IZ} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ \text{XI} & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \text{XX} & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \text{XY} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ \text{XZ} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 \\ \text{YI} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ \text{YX} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \text{YY} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \text{YZ} & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \text{ZI} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ \text{ZX} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ \text{ZY} & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \text{ZZ} & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

$$\chi = \begin{pmatrix} & \text{II} & \text{IX} & \text{IY} & \text{IZ} & \text{XI} & \text{XX} & \text{XY} & \text{XZ} & \text{YI} & \text{YX} & \text{YY} & \text{YZ} & \text{ZI} & \text{ZX} & \text{ZY} & \text{ZZ} \\ \text{II} & \frac{1}{4} & \frac{1}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{4} & -\frac{1}{4} & 0 & 0 \\ \text{IX} & \frac{1}{4} & \frac{1}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{4} & -\frac{1}{4} & 0 & 0 \\ \text{IY} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \text{IZ} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \text{XI} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \text{XX} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \text{XY} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \text{XZ} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \text{YI} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \text{YX} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \text{YY} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \text{YZ} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \text{ZI} & \frac{1}{4} & \frac{1}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{4} & -\frac{1}{4} & 0 & 0 \\ \text{ZX} & -\frac{1}{4} & -\frac{1}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{4} & \frac{1}{4} & 0 & 0 \\ \text{ZY} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \text{ZZ} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$



Yes

27.5 Pauli conjugated by Pauli [MOVE]

Aim I would like to calculate the very useful construction where we conjugate a Pauli P_b either on the control, target, or on both, by the control-Pauli. Let's work these out one by one.